

Phase I/II Operating Characteristics

Toxicity-prior comparison - two-stage - IPDE alpha = 0 - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	30.4	11	1	0.4	0.2	57
Toxicity prior Beta(0.3, 0.7)	31.3	10.8	1.3	0.8	0.2	55.6
Toxicity prior Beta(0.5, 0.5)	32.8	10.5	2.5	0.3	0.2	53.6
Toxicity prior Beta(1, 1)	32.9	9.7	2.8	0.6	0.2	53.8
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	15.4	3.8	2.1	0.9	0.3	
Toxicity prior Beta(0.3, 0.7)	15.2	3.9	2.2	1	0.4	
Toxicity prior Beta(0.5, 0.5)	14.8	4.2	2.4	1.1	0.5	
Toxicity prior Beta(1, 1)	14.7	4.2	2.4	1.1	0.5	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	34.8	16.2	32	11.9	3.2	1.9
Toxicity prior Beta(0.3, 0.7)	34.3	14.9	32.1	12.4	4.4	1.8
Toxicity prior Beta(0.5, 0.5)	31.9	14.1	32.4	14.7	5.1	1.8
Toxicity prior Beta(1, 1)	31.1	14.6	32.7	14.7	5.1	1.8
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	11.4	5.2	7.2	3.9	2	
Toxicity prior Beta(0.3, 0.7)	11.3	5	7.1	3.9	2.5	
Toxicity prior Beta(0.5, 0.5)	10.9	4.7	7	4.4	2.7	
Toxicity prior Beta(1, 1)	10.9	4.7	7	4.5	2.7	
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	18.4	40.3	21.4	7.3	2.6	10
Toxicity prior Beta(0.3, 0.7)	17.4	38.9	22.5	9	2.3	9.8
Toxicity prior Beta(0.5, 0.5)	16.4	38.1	22.5	9.9	3.4	9.7
Toxicity prior Beta(1, 1)	16.3	37.6	22.8	10.1	3.4	9.8
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	9.7	7.5	6.2	3.7	1.9	
Toxicity prior Beta(0.3, 0.7)	9.6	7.3	6.1	3.7	2.2	
Toxicity prior Beta(0.5, 0.5)	9.2	7	6.2	4.1	2.5	
Toxicity prior Beta(1, 1)	9.3	7	6.1	4.1	2.5	

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	7.3	8.3	21.3	42.1	19.1	1.9
Toxicity prior Beta(0.3, 0.7)	7.1	7	20.2	43	20.8	1.8
Toxicity prior Beta(0.5, 0.5)	5.6	6.2	18.1	44.8	23.5	1.8
Toxicity prior Beta(1, 1)	5.5	5.5	19.1	44.9	23.1	1.8
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	4.5	4.3	6.6	7.9	6.4	
Toxicity prior Beta(0.3, 0.7)	4.4	4.1	6.1	8.2	6.9	
Toxicity prior Beta(0.5, 0.5)	4.2	3.8	5.7	8.7	7.2	
Toxicity prior Beta(1, 1)	4.2	3.8	5.7	8.8	7.2	
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	5	34.7	26.7	18.8	10.5	4.3
Toxicity prior Beta(0.3, 0.7)	4.8	35.2	25.4	18.3	12.1	4.2
Toxicity prior Beta(0.5, 0.5)	4.8	34.3	26	18.6	12	4.2
Toxicity prior Beta(1, 1)	4.8	34.8	25.5	18.6	12	4.3
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	5	6.5	6.6	5.8	5.5	
Toxicity prior Beta(0.3, 0.7)	4.9	6.4	6.5	5.7	5.9	
Toxicity prior Beta(0.5, 0.5)	4.9	6.2	6.5	5.7	6.2	
Toxicity prior Beta(1, 1)	4.9	6.1	6.6	5.7	6.1	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	68.2	12.9	3.6	0.9	0.6	13.8
Toxicity prior Beta(0.3, 0.7)	67.1	13.3	4.2	1.6	0.4	13.3
Toxicity prior Beta(0.5, 0.5)	64.8	14.4	5	1.9	0.6	13.4
Toxicity prior Beta(1, 1)	64.8	14	5.5	1.9	0.5	13.3
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	19.5	4	2.5	1.1	0.4	
Toxicity prior Beta(0.3, 0.7)	19.4	3.9	2.6	1.2	0.5	
Toxicity prior Beta(0.5, 0.5)	18.8	4.1	2.7	1.4	0.6	
Toxicity prior Beta(1, 1)	18.8	4.2	2.6	1.4	0.6	

Phase I/II Operating Characteristics

Toxicity-prior comparison - two-stage - IPDE alpha = 0 - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	26.2	36.7	20.3	10.5	4	2.3
Toxicity prior Beta(0.3, 0.7)	26.2	35.8	21.4	9.5	4.9	2.2
Toxicity prior Beta(0.5, 0.5)	25.4	33.9	21.5	11.6	5.3	2.2
Toxicity prior Beta(1, 1)	25.1	34.8	20.8	11.3	5.6	2.2
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	10.8	7	5.6	3.7	2.4	
Toxicity prior Beta(0.3, 0.7)	10.7	6.9	5.6	3.7	2.8	
Toxicity prior Beta(0.5, 0.5)	10.3	6.5	5.7	4	3.1	
Toxicity prior Beta(1, 1)	10.3	6.6	5.6	4.1	3.1	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	43.3	9.3	2	1.2	0.4	43.8
Toxicity prior Beta(0.3, 0.7)	42.2	10.1	2.9	1.2	0.4	43.2
Toxicity prior Beta(0.5, 0.5)	41.6	10.5	4	1.3	0.7	41.9
Toxicity prior Beta(1, 1)	41.7	9.4	3.9	1.7	0.8	42.5
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	17.4	3.7	1.8	0.8	0.3	
Toxicity prior Beta(0.3, 0.7)	17.2	3.7	1.9	0.9	0.3	
Toxicity prior Beta(0.5, 0.5)	16.6	4	2.1	1	0.4	
Toxicity prior Beta(1, 1)	16.6	4	2.1	1	0.4	
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	13.6	10.7	16.9	13.5	26	19.3
Toxicity prior Beta(0.3, 0.7)	12.8	9.4	17.1	14.2	27.9	18.5
Toxicity prior Beta(0.5, 0.5)	12.7	8.5	16.1	13.9	30.6	18.3
Toxicity prior Beta(1, 1)	11.6	8.7	16.8	12.6	31.9	18.5
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	8.9	4.3	5.4	4	5.1	
Toxicity prior Beta(0.3, 0.7)	8.8	4.1	5.2	4	5.7	
Toxicity prior Beta(0.5, 0.5)	8.4	4	5	4.3	6.2	
Toxicity prior Beta(1, 1)	8.4	4	4.9	4.4	6.2	

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Toxicity-prior comparison - two-stage - IPDE alpha = 0 - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	11.7	31.1	12.4	4.3	2	38.5
Toxicity prior Beta(0.3, 0.7)	11.2	30.1	14.1	4.5	2.5	37.6
Toxicity prior Beta(0.5, 0.5)	10.9	28	14	6.9	3.3	36.9
Toxicity prior Beta(1, 1)	10.6	27.6	14.9	6.3	3.6	37
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	11.8	5.8	4.2	2.1	1.1	
Toxicity prior Beta(0.3, 0.7)	11.6	5.7	4.2	2.3	1.4	
Toxicity prior Beta(0.5, 0.5)	11	5.9	4.2	2.7	1.6	
Toxicity prior Beta(1, 1)	11	6	4.1	2.6	1.6	
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	66.6	21.4	4.1	0.7	0	7.2
Toxicity prior Beta(0.3, 0.7)	65.5	22.8	4.2	0.7	0	6.7
Toxicity prior Beta(0.5, 0.5)	65.7	21.3	5.5	0.8	0.1	6.5
Toxicity prior Beta(1, 1)	65	23.1	4.8	0.8	0	6.3
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	18.5	5.8	3.5	1	0.1	
Toxicity prior Beta(0.3, 0.7)	18.2	5.8	3.6	1.2	0.2	
Toxicity prior Beta(0.5, 0.5)	17.5	6	3.6	1.5	0.3	
Toxicity prior Beta(1, 1)	17.5	6.1	3.6	1.5	0.3	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	15.1	12.2	9.4	36.3	23.8	3.2
Toxicity prior Beta(0.3, 0.7)	15.1	10.9	9.3	36.7	24.7	3.2
Toxicity prior Beta(0.5, 0.5)	12.1	10	7.6	38	29.1	3.2
Toxicity prior Beta(1, 1)	11.4	8.3	8.2	38.6	30.2	3.3
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	5.1	4.6	5.3	7.4	7.1	
Toxicity prior Beta(0.3, 0.7)	4.9	4.4	4.9	7.6	7.7	
Toxicity prior Beta(0.5, 0.5)	4.7	4.1	4.7	7.9	8.2	
Toxicity prior Beta(1, 1)	4.7	4.1	4.6	8	8.2	

Phase I/II Operating Characteristics

Toxicity-prior comparison - two-stage - IPDE alpha = 0 - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	13.4	33.3	13.3	4.3	1.8	33.9
Toxicity prior Beta(0.3, 0.7)	12.9	32.5	14.8	5.4	1.9	32.5
Toxicity prior Beta(0.5, 0.5)	12.7	30.3	15.6	7.4	2.2	31.8
Toxicity prior Beta(1, 1)	12.2	29.9	16.6	6.8	2.7	31.8
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	11.5	6.2	4.5	2.4	1.2	
Toxicity prior Beta(0.3, 0.7)	11.2	6	4.6	2.5	1.4	
Toxicity prior Beta(0.5, 0.5)	10.5	6.3	4.7	2.9	1.5	
Toxicity prior Beta(1, 1)	10.5	6.2	4.7	2.9	1.6	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	20.7	14	13	29.6	15.3	7.4
Toxicity prior Beta(0.3, 0.7)	19.6	13.1	12.2	29.1	18.6	7.3
Toxicity prior Beta(0.5, 0.5)	17.9	12.9	10.1	31.5	20.3	7.3
Toxicity prior Beta(1, 1)	17.1	12.4	10.4	32.4	20.6	7.2
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	6.8	5.3	5.8	6.1	5	
Toxicity prior Beta(0.3, 0.7)	6.6	5.1	5.5	6.3	5.5	
Toxicity prior Beta(0.5, 0.5)	6.2	4.8	5.3	6.8	6	
Toxicity prior Beta(1, 1)	6.3	4.7	5.3	6.9	6	
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	54.2	23.6	10.1	3.7	1.3	7.1
Toxicity prior Beta(0.3, 0.7)	54.4	23	10.2	4.4	1	6.9
Toxicity prior Beta(0.5, 0.5)	53.7	21.5	12.4	4.5	0.7	7.3
Toxicity prior Beta(1, 1)	53	23.2	11.8	4.3	0.5	7.2
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	11.7	6.7	5.8	3.4	1.6	
Toxicity prior Beta(0.3, 0.7)	11.5	6.5	5.8	3.4	1.9	
Toxicity prior Beta(0.5, 0.5)	11.1	6.2	5.7	4	2.1	
Toxicity prior Beta(1, 1)	11.1	6.3	5.7	4	2.2	

Phase I/II Operating Characteristics

Toxicity-prior comparison - two-stage - IPDE alpha = 0 - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	30.6	23.5	16	17.2	12.6	0.1
Toxicity prior Beta(0.3, 0.7)	29.6	24.6	16.8	15.6	13.2	0.1
Toxicity prior Beta(0.5, 0.5)	32	23.1	15.5	16	13.3	0.1
Toxicity prior Beta(1, 1)	30.2	23.2	16.4	16.3	13.7	0.1
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	6	5.5	5.5	5.9	7.2	
Toxicity prior Beta(0.3, 0.7)	5.9	5.5	5.4	5.7	7.4	
Toxicity prior Beta(0.5, 0.5)	5.9	5.4	5.4	5.7	7.6	
Toxicity prior Beta(1, 1)	5.9	5.4	5.5	5.7	7.5	
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	31.2	49.6	17	1.9	0	0.3
Toxicity prior Beta(0.3, 0.7)	29.8	49	18.6	2.3	0	0.3
Toxicity prior Beta(0.5, 0.5)	29.9	48.3	18.8	2.7	0	0.3
Toxicity prior Beta(1, 1)	29.8	48	18.8	3.1	0	0.3
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	10.3	9.7	7.2	2.3	0.4	
Toxicity prior Beta(0.3, 0.7)	10.2	9.4	7.3	2.6	0.5	
Toxicity prior Beta(0.5, 0.5)	9.5	9.3	7.4	3.1	0.6	
Toxicity prior Beta(1, 1)	9.5	9.2	7.4	3.1	0.6	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	11.6	6.5	9.5	13.8	57.9	0.7
Toxicity prior Beta(0.3, 0.7)	11	7.1	8.1	13.1	59.9	0.7
Toxicity prior Beta(0.5, 0.5)	8.5	7.6	8	12.8	62.5	0.7
Toxicity prior Beta(1, 1)	8.4	7.2	8.5	12.2	63.1	0.7
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	3.7	3.9	4.7	5.7	11.9	
Toxicity prior Beta(0.3, 0.7)	3.7	3.8	4.4	5.4	12.5	
Toxicity prior Beta(0.5, 0.5)	3.7	3.8	4.3	5.4	12.7	
Toxicity prior Beta(1, 1)	3.7	3.8	4.3	5.5	12.6	

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	19.9	52.8	17.2	1.5	0.1	8.5
Toxicity prior Beta(0.3, 0.7)	19.1	52.7	17.8	2.2	0	8.1
Toxicity prior Beta(0.5, 0.5)	19.5	48.9	21.6	1.9	0.2	8
Toxicity prior Beta(1, 1)	19	49.6	20.6	2.5	0.2	8.1
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	9.1	10.3	7	2.2	0.3	
Toxicity prior Beta(0.3, 0.7)	9	9.9	7.1	2.5	0.4	
Toxicity prior Beta(0.5, 0.5)	8.3	9.9	7.3	3.1	0.5	
Toxicity prior Beta(1, 1)	8.4	9.7	7.4	3.1	0.5	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	12.1	45.3	27.3	9.9	1.2	4.2
Toxicity prior Beta(0.3, 0.7)	12.4	43.9	28.2	9.9	1.4	4.2
Toxicity prior Beta(0.5, 0.5)	12.7	41.6	27.5	11.6	2.3	4.2
Toxicity prior Beta(1, 1)	11.2	41.6	27.5	12.9	2.6	4.3
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	5.7	8.7	8.3	4.5	2.2	
Toxicity prior Beta(0.3, 0.7)	5.5	8.3	8.1	5	2.4	
Toxicity prior Beta(0.5, 0.5)	5.3	7.8	7.8	5.8	2.7	
Toxicity prior Beta(1, 1)	5.3	7.8	7.9	5.8	2.7	
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	60.1	1.4	0	0	0	38.5
Toxicity prior Beta(0.3, 0.7)	59.9	1.9	0	0	0	38.2
Toxicity prior Beta(0.5, 0.5)	60.2	1.9	0	0	0	37.9
Toxicity prior Beta(1, 1)	60.3	1.8	0	0	0	37.9
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	20.8	2.1	0.6	0	0	
Toxicity prior Beta(0.3, 0.7)	20.9	2.1	0.6	0	0	
Toxicity prior Beta(0.5, 0.5)	20.5	2.4	0.7	0.1	0	
Toxicity prior Beta(1, 1)	20.5	2.4	0.7	0.1	0	

Phase I/II Operating Characteristics

Toxicity-prior comparison - two-stage - IPDE alpha = 0 - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	26.9	55.4	11.6	1.6	0	4.5
Toxicity prior Beta(0.3, 0.7)	25.9	54.2	13.3	2.5	0	4.1
Toxicity prior Beta(0.5, 0.5)	23.9	53.5	15.2	3.1	0	4.2
Toxicity prior Beta(1, 1)	24.2	52.7	15.3	3.5	0	4.3
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	8.7	12.2	6.5	1.7	0.2	
Toxicity prior Beta(0.3, 0.7)	8.6	11.8	6.7	1.9	0.4	
Toxicity prior Beta(0.5, 0.5)	7.9	11.8	6.6	2.7	0.4	
Toxicity prior Beta(1, 1)	8	11.5	6.7	2.8	0.5	
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	19.7	5.2	0.2	0	0	74.9
Toxicity prior Beta(0.3, 0.7)	20.9	5.8	0.1	0	0	73.1
Toxicity prior Beta(0.5, 0.5)	22.1	4.8	0.2	0	0	72.9
Toxicity prior Beta(1, 1)	22.5	5.1	0.1	0	0	72.3
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	17.2	2.7	0.6	0	0	
Toxicity prior Beta(0.3, 0.7)	17.2	2.8	0.6	0	0	
Toxicity prior Beta(0.5, 0.5)	17	3.1	0.7	0.1	0	
Toxicity prior Beta(1, 1)	17	3.1	0.7	0.1	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	1	1.7	4.9	44.8	47.5	0.1
Toxicity prior Beta(0.3, 0.7)	1	1.8	5.2	44.4	47.5	0.1
Toxicity prior Beta(0.5, 0.5)	1	1.3	5.5	44	48.2	0
Toxicity prior Beta(1, 1)	1	1.4	5.5	43.9	48.1	0
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	3.3	3.3	3.8	8.3	11.3	
Toxicity prior Beta(0.3, 0.7)	3.3	3.3	3.7	8.2	11.5	
Toxicity prior Beta(0.5, 0.5)	3.3	3.2	3.8	8.2	11.5	
Toxicity prior Beta(1, 1)	3.3	3.2	3.8	8.1	11.6	

Phase I/II Operating Characteristics

Toxicity-prior comparison - two-stage - IPDE alpha = 0 - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	35.5	28	31	4.9	0	0.6
Toxicity prior Beta(0.3, 0.7)	34.3	27.4	32.1	5.5	0.1	0.7
Toxicity prior Beta(0.5, 0.5)	34.4	24.5	34.7	5.7	0	0.6
Toxicity prior Beta(1, 1)	33.6	25.4	34.4	5.9	0	0.6
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	9.4	8	9.3	2.7	0.5	
Toxicity prior Beta(0.3, 0.7)	9.1	7.7	9.4	3.1	0.6	
Toxicity prior Beta(0.5, 0.5)	8.8	7.1	9.4	3.9	0.7	
Toxicity prior Beta(1, 1)	8.7	7.1	9.4	3.9	0.7	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	28.8	28.3	27.4	5.2	0.1	10.2
Toxicity prior Beta(0.3, 0.7)	28.3	26.9	28.1	6.8	0.1	9.9
Toxicity prior Beta(0.5, 0.5)	27.6	22.9	31.2	8	0.1	10.2
Toxicity prior Beta(1, 1)	27.3	24	30.1	8.2	0.2	10.2
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	10.9	6.7	7.9	2.8	0.5	
Toxicity prior Beta(0.3, 0.7)	10.7	6.4	8	3.2	0.6	
Toxicity prior Beta(0.5, 0.5)	10.3	6	8.3	3.7	0.7	
Toxicity prior Beta(1, 1)	10.3	5.9	8.2	3.8	0.7	
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	22	21.7	41.8	8.1	0.9	5.5
Toxicity prior Beta(0.3, 0.7)	20.1	20.8	41.9	10.7	1.1	5.3
Toxicity prior Beta(0.5, 0.5)	20.2	17.6	44.7	10.8	1.2	5.5
Toxicity prior Beta(1, 1)	20.1	17.9	42.9	12.3	1.4	5.3
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	8.2	6.4	9.4	4	1.3	
Toxicity prior Beta(0.3, 0.7)	8.1	6.1	9.2	4.3	1.6	
Toxicity prior Beta(0.5, 0.5)	7.7	5.6	9.3	5.1	1.7	
Toxicity prior Beta(1, 1)	7.7	5.6	9.2	5.3	1.7	

Phase I/II Operating Characteristics

Toxicity-prior comparison - two-stage - IPDE alpha = 0 - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	21.3	38	6.5	0.2	0	34
Toxicity prior Beta(0.3, 0.7)	20.5	38.5	6.7	0.8	0	33.5
Toxicity prior Beta(0.5, 0.5)	21.1	39	6.9	1.2	0	31.7
Toxicity prior Beta(1, 1)	20.4	39.3	7.6	0.9	0	31.8
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	12.1	8.4	4.2	0.9	0.1	
Toxicity prior Beta(0.3, 0.7)	12	8.2	4.5	1	0.2	
Toxicity prior Beta(0.5, 0.5)	11.1	8.9	4.4	1.4	0.2	
Toxicity prior Beta(1, 1)	11.2	8.7	4.5	1.5	0.2	
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	68.6	24.5	4	0.3	0	2.6
Toxicity prior Beta(0.3, 0.7)	69.2	23.6	4.3	0.1	0	2.7
Toxicity prior Beta(0.5, 0.5)	68.3	25	4.3	0.1	0	2.3
Toxicity prior Beta(1, 1)	67.5	25.3	4.5	0.2	0	2.6
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	18.2	7.2	3.5	0.5	0	
Toxicity prior Beta(0.3, 0.7)	18	7	3.7	0.6	0.1	
Toxicity prior Beta(0.5, 0.5)	17.4	7.5	3.8	0.9	0.1	
Toxicity prior Beta(1, 1)	17.2	7.5	3.8	0.9	0.1	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	21.1	13.8	15.5	36.4	12.4	0.8
Toxicity prior Beta(0.3, 0.7)	21.5	12.3	15.6	36.7	13	0.8
Toxicity prior Beta(0.5, 0.5)	17.9	11.1	12.9	41.5	15.8	0.8
Toxicity prior Beta(1, 1)	17.6	11.5	11.7	42.2	16.3	0.8
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	5.2	4.6	5.8	8.2	6.2	
Toxicity prior Beta(0.3, 0.7)	5.2	4.4	5.5	8.1	6.8	
Toxicity prior Beta(0.5, 0.5)	5	4.2	5.3	8.5	6.9	
Toxicity prior Beta(1, 1)	4.9	4.2	5.2	8.6	7	

Phase I/II Operating Characteristics

Toxicity-prior comparison - two-stage - IPDE alpha = 0 - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	7	36.1	26	24.3	6.5	0.1
Toxicity prior Beta(0.3, 0.7)	6.9	35.6	24.9	25.3	7.2	0.1
Toxicity prior Beta(0.5, 0.5)	7	34.2	24.5	24.8	9.5	0.1
Toxicity prior Beta(1, 1)	7.4	35.5	24.8	23.7	8.5	0.1
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	4.1	5.9	6.5	7.6	5.9	
Toxicity prior Beta(0.3, 0.7)	4.1	5.7	6.3	7.5	6.3	
Toxicity prior Beta(0.5, 0.5)	4.1	5.7	6.2	7.6	6.4	
Toxicity prior Beta(1, 1)	4.1	5.7	6.2	7.5	6.5	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	9.5	13.6	33.8	28	14.5	0.6
Toxicity prior Beta(0.3, 0.7)	9.1	13.1	34.6	27.7	14.9	0.6
Toxicity prior Beta(0.5, 0.5)	8.7	14.1	34.8	27.9	14	0.6
Toxicity prior Beta(1, 1)	8.3	13.8	33.4	29.1	14.7	0.6
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	4	4.3	7.7	7.2	6.7	
Toxicity prior Beta(0.3, 0.7)	4	4.3	7.4	7.1	7.2	
Toxicity prior Beta(0.5, 0.5)	4	4.2	7.3	7.1	7.3	
Toxicity prior Beta(1, 1)	4	4.2	7.2	7.3	7.3	
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	9.1	4.5	12.8	40.5	33.1	0
Toxicity prior Beta(0.3, 0.7)	8.3	4.4	12.6	39.3	35.4	0
Toxicity prior Beta(0.5, 0.5)	7.3	3.8	11	39.3	38.5	0
Toxicity prior Beta(1, 1)	7.2	4.6	11.1	38.7	38.4	0
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	3.5	3.6	5.3	8.1	9.4	
Toxicity prior Beta(0.3, 0.7)	3.5	3.6	5	7.9	10	
Toxicity prior Beta(0.5, 0.5)	3.4	3.5	4.7	8.3	10	
Toxicity prior Beta(1, 1)	3.4	3.5	4.7	8.3	10.1	

Phase I/II Operating Characteristics

Toxicity-prior comparison - two-stage - IPDE alpha = 0 - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	14.7	12.1	47.6	18.8	4.8	2
Toxicity prior Beta(0.3, 0.7)	13.4	11.2	46.9	19.9	6.5	2
Toxicity prior Beta(0.5, 0.5)	11.1	10.9	44.2	23.9	8	1.9
Toxicity prior Beta(1, 1)	11.6	11.1	43.4	23.5	8.5	1.9
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	5.3	5.8	10	5.5	3.2	
Toxicity prior Beta(0.3, 0.7)	5.2	5.4	9.8	5.5	3.9	
Toxicity prior Beta(0.5, 0.5)	4.9	5.1	9.1	6.4	4.3	
Toxicity prior Beta(1, 1)	4.9	5.1	9.1	6.4	4.3	
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	11.7	40.5	31.4	11.8	4.5	0.1
Toxicity prior Beta(0.3, 0.7)	11.2	39.3	31.8	12.9	4.7	0.1
Toxicity prior Beta(0.5, 0.5)	10.3	39.1	31.2	14.5	4.9	0.1
Toxicity prior Beta(1, 1)	10.1	39.1	29.1	15.7	5.8	0.1
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	4.6	8.4	8.9	5.1	3	
Toxicity prior Beta(0.3, 0.7)	4.6	8	8.7	5.1	3.5	
Toxicity prior Beta(0.5, 0.5)	4.5	7.5	8.3	5.9	3.7	
Toxicity prior Beta(1, 1)	4.5	7.6	8.1	6	3.7	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	20.9	41.6	14.4	6	1.7	15.4
Toxicity prior Beta(0.3, 0.7)	19.6	38.8	16.8	7.6	2.3	14.8
Toxicity prior Beta(0.5, 0.5)	17.2	38.9	17.5	9	3.2	14.2
Toxicity prior Beta(1, 1)	18.3	37.4	17.3	9.3	3.2	14.5
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	9.7	8.3	5.8	2.8	1.3	
Toxicity prior Beta(0.3, 0.7)	9.5	8	5.8	3	1.6	
Toxicity prior Beta(0.5, 0.5)	8.7	8.2	5.8	3.6	1.9	
Toxicity prior Beta(1, 1)	8.8	8.1	5.7	3.6	1.9	
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
Toxicity prior Beta(0.15, 0.85)	27.2	48.4	13.1	3.4	0.7	7.2
Toxicity prior Beta(0.3, 0.7)	25.5	50	13.4	3.5	0.7	6.9
Toxicity prior Beta(0.5, 0.5)	24.6	47.7	15.4	4.3	1.2	6.8
Toxicity prior Beta(1, 1)	23.5	48.6	15	5	1.1	6.7
Mean patients treated						
Toxicity prior Beta(0.15, 0.85)	8.7	10.7	6.2	2.3	0.8	
Toxicity prior Beta(0.3, 0.7)	8.5	10.4	6.2	2.5	1.1	
Toxicity prior Beta(0.5, 0.5)	7.9	10.5	6.2	3.1	1.2	
Toxicity prior Beta(1, 1)	7.8	10.4	6.3	3.1	1.2	